EXTENDING VISUOSPATIAL ANALYSIS IN DESIGN COMPUTING

An Exploration with a Novel GPU-Based Algorithm and Form-Based Codes

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Abstract. This paper responds to a gap observed between the contemporary capacity for calculation and analysis of visibility of built environment features, such as buildings, in digital urban and architectural computational research models and the functionality of off-the-shelf software tools available to professionals. The research investigates the potential of visibility analysis to be embedded and extended within computational-based workflows of software tools to better meet urban design and planning industry needs. We introduce a novel method for visibility calculation that exposes output data for further analysis within a computational workflow and implement it in a game development engine used by software tool providers. Based in our engagement with a local government authority, we then use that method to demonstrate a workflow in the context of form-based building codes in which the visual impact of a building is considered rather than prescriptive limits on dimensions and use. Our results indicate the novel method has substantial performance improvements compared to an alternative mode of visibility calculation and that software providers could more thoroughly integrate and extend visibility analysis to meet industry needs.

Keywords. Design Computing, Viewsheds, Isovists, GPU Shader, Unity 3D, Genetic Algorithm, Generative Design, Form-based Building Codes

1. Introduction

Built environment design computing workflows that support visibility analysis, if well implemented, can be an invaluable tool in urban densification design and planning. Growing urban populations place pressures on the resources and infrastructure of cities and the capacities of those responsible for their planning, design, and maintenance (United Nations 2019). The consequent urban densification, if managed skilfully, can contribute positively to social well-being through efficiencies of scale, improved sustainability, and health outcomes (Buxton 2014;

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Trubka et al. 2010). Aspects of good urban densification design and planning rely on the understanding of visuospatial influences of urban features, such as form-based building codes, in which the visual impact of a building is considered rather than stringent limits on use and dimensions, or urban greening strategies (Talen 2013; Yang et al. 2009). This paper questions how design computing workflows may efficiently and accessibly incorporate and extend urban visibility analysis as an integral component of the urban design process to address real-world problems.

2. Background and Context

The fundamental element of urban visibility analysis is the relationship between a feature of interest – an attractive park or a selected building, perhaps - and a location from which its visibility is to be tested – possibly a point on a building façade or a vantage point on a nearby street, correspondingly. Notably, that relationship between the feature and location is 1D and is, in effect, a hit test of a notional ray extending from the viewpoint location toward where the feature of interest may (or may not) exist. That test, although 1D, may be oriented in 2D, 3D, or, as 2.5D, pseudo-3D space; it, in its sweep through these higher dimensions, becomes a viewshed (Figure 1).

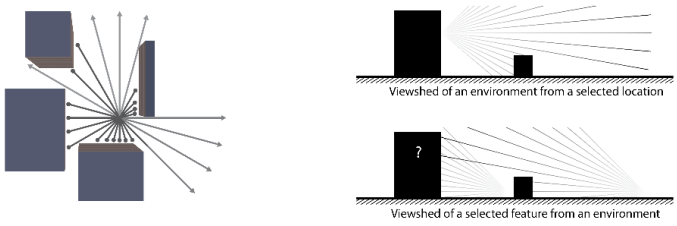


Figure 1. Conceptual diagrams of a 2D viewshed (left) and viewsheds from a location and of a feature (right).

It is that latter pseudo-3D viewshed analysis that has been prevalent in visuospatial computing workflows. Early work focused on calculating viewsheds in digital elevation models (DEM) in which each cell or raster pixel represents a height value, beginning with methods for which "preparing terrain data in computer-readable form" would "usually be punched cards" (Travis 1975). This 2.5D approach has continued with a succession of improvements on algorithms and data structures to lead to more efficient calculation, including with methods that exploit the parallel processing of the graphics processing unit (GPU) (Zhao et al. 2013). The DEM method is suited to terrain analysis that is not impeded by the 2.5D modelling limitation that no surface may exist above any other.

Other methods apply in 2D space, typically for the architectural plan, in which the viewshed is calculated as a bounding polygon, akin to what the protagonist of Abbott's 1884 Flatland: A Romance of Many Dimensions would experience of his 2D universe. The essential concept of this polygon, or isovist, has been extended to statistical measures of perceptual qualities – at the most elementary, area and perimeter length – in 2D and 3D, and isovist fields in which measures taken throughout the space of an environment can be mapped as a scalar field or heatmap-

style visualisation (Benedict 1979, Batty 2001).

Contemporary advances in computing power and the take up of 3D modelling in BIM, PIM, and other workflows are strengthening the case for effective 3D approaches. Extending the quantised modelling of DEMs, Morello and Ratti (2009) efficiently computed 2.5D isovists within a 3D voxel grid and evaluated their application to quantitative indicators of visual elements. For vertex-based 3D modelling, the parallel computing of GPUs is used to efficiently calculate viewsheds by Chao et al. (2011) and others, and 3D isovists have been explored (Derix et al. 2008). Adapting tools from outside the AEC industry to urban issues, White and Langenheim (2014) explored the appropriation of light sources within 3D animation software to assess view quality from buildings.

Visuospatial modelling as described above typically stops at an immediate stage of presentation of visualisation of viewsheds or their meta-analysis, at which the design computing workflow halts and the human is brought into the design loop. Users can benefit from this visual analysis and apply their understanding of urban environments as complex systems and of the nuanced considerations of real-world projects – that may not easily be accommodated in design computing algorithms – to further iterations of the design loop. Might not, though, the human-in-the-loop benefit also from extending further the computation on visuospatial analysis outputs in that same design loop?

Some approaches do propagate analysis results further within the design computing workflow. Viewshed analysis in 2D and 3D has been linked with agent-based modelling (ABM) to simulate pedestrian movement with respect to experiential qualities of line of sight, spatial openness, or visual occlusion (Aschwanden et al. 2008; depthmapX development team 2017). Akin to ABM, Lee and Stucky (1998) employed viewsheds within a DEM to calculate least-cost paths that maximise or minimise their visibility from or of non-path DEM cells. Via the DecodingSpaces Toolbox (toolbox.decodingspaces.net), viewshed analysis has been embedded within the Grasshopper visual scripting environment to allow visible entities to be explicitly identified and for visibility data to be passed further along the computational design workflow of a Grasshopper script. In 2.5D, viewshed analysis has been adapted to optimise precinct building envelopes for solar amenity as experienced by ABM pedestrians (Kimm 2020). In other generative work, Schneider and König (2012) applied computational optimisation to diverse isovist field statistical measures to generate building footprints.

However, a review we conducted of leading commercial precinct modelling software platforms targeted to design and planning professionals indicates that this data extensibility does not typically carry over to off-the-shelf tools available to industry. Spacemaker (autodesk.com/spacemaker), an Autodesk product for designing site development proposals, permits users to undertake view analysis as either view distance – a measure of how distant the view is at a point on a façade – or view to area – a measure of whether an area of interest is visible from a given façade point. As with the view to area of Spacemaker, CityEngine (esri.com/cityengine) supports the calculation of viewsheds for which the visibility information from a point, and within a limited field of view, is projected onto the urban model. That visibility information – default red for hidden and green for visible – may then be

viewed by the user from other perspectives within the model. The planning and design urban digital twin application VU.CITY (www.vu.city) provides viewshed analysis as *zone of theoretical visibility* in which ground-level analysis of the visibility of a selected building is undertaken. The heatmap-style visualisation provides a colour scale corresponding to the visible vertical proportion of the building. Unsigned Studio (unsignedstudio.com), a division of the international engineering, design, and advisory firm Aurecon, produces siteLab, a built environment visualisation tool that operates from precinct level to the scale of architectural detail, that provides what-if view frustum testing of CCTV camera installation for public safety. Many geographic information system (GIS) packages can perform viewshed analysis. For example, GRASS GIS (grass.osgeo.org) can calculate an output image of which modelling cells are visible from a selected point on a raster terrain.

While these representative industry examples are varied in their application of visibility analysis, their use has characteristics in common. They are highly graphically based analyses that are primarily oriented to visual display of data within the 3D scene. Although some like CityEngine, which includes a breakdown of the share of layer visibility and like interpretation, go further, information typically is focused toward human- rather than machine-oriented additional processing.

Our research therefore considers how visuospatial analysis may be embedded within a design computing workflow in which its outputs may propagate further, and how this may be implemented in a way accessible and relevant to industry software solution providers. It first presents a novel GPU-based algorithm demonstrated in a game development engine. Using that algorithm, it then examines a context developed in collaboration with a local government authority of outer Melbourne, Australia, to test extending visuospatial analysis in a design computing workflow.

3. Method

A novel urban visibility algorithm was developed in the Unity 3D game engine ("Unity") to efficiently expose viewshed analysis for further use in a design computing workflow. The algorithm was then tested in a real-world context.

The GPU-based urban visibility analysis algorithm extends two precedents. Schneider et al. (2014) used a GPU-based method for real-time solar analysis that can assess performance criteria at the early design stage. In their method, a cube map is rendered for a given point within a digital 3D urban model against a sky image that embodies radiation values for a single instant in time or a specified period. The illuminance and daylight factor at the given point can be calculated from the rendered image by summating the intensity values of the sky that is unoccluded by the massed buildings of the 3D model. Chirkin et al. (2018) extended the approach of Schneider et al. to evaluation of massed objects of a 3D urban scene directly, and demonstrated the generalised applicability of their approach to analyses including distance to the closest object and visible ground area. In the methods of Schneider et al. and of Chirkin et al., evaluations may be taken across a multitude of points to display a heatmap-style precinct-level visualisation.

The adaptation of the 3D GPU pipeline of the precedents was extended in an

algorithm for viewshed analysis of individual programmatically specified urban features and their occlusion by their context; through this flexibility, it is extensible to experiential perspectives within the urban scene rather than simply global or static perspectives. Visibility analysis data of a specified feature is exposed to subsequent stages of a design computing workflow.

The algorithm is a workflow of six key steps implemented in this research in the Unity 3D game engine ("Unity") (Figure 2).

- Two cameras, A and B, are instantiated in the scene and positioned to view the subject building from a desired location. Camera A is set to clear a shared target render texture to a predefined background colour and to render only the subject building. Camera B is set to render any obscuring context.
- A shader, being a small program that instructs a GPU how to draw parts of the scene, uses camera A to render the building with a gradient according to its height.
- A second shader on camera B renders all context with a predefined context colour.
- A compute shader, being a program that allows user-defined parallel data processing on the GPU, accepts the shared render texture and calculates a histogram and other statistical measures of the visibility of the rendered building.
- The output of the compute shader is waited on asynchronously and is added to a queue data structure when available.
- Results read from the queue are processed per application requirements.

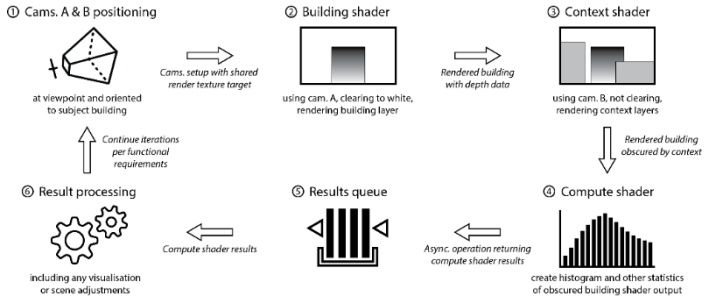


Figure 2. The algorithm of the viewshed calculation as implemented in Unity.

Unity was selected as an implementation platform for its industry acceptance, accessibility, and broad use in built environment analysis and simulation contexts (Huang et al. 2021). It is utilised in the AEC and urban design and planning industries by platforms including VU.CITY and siteLAB, each discussed above, and offers the Unity Reflect product suite of BIM AR and VR with connection to Autodesk Revit (unity.com/products/unity-reflect).

The application of the algorithm for exploiting viewshed analysis outputs in stages of the design workflow beyond immediate visualisation was tested. For this, a test context was identified in consultation with an outer Melbourne local government authority ("the LGA"). The results were then analysed for performance.

4. Results

The test context of a "serious gap" was identified in discussion with the LGA. "One of the challenges we have in [planning and design for] an urban context" is that "often everything is perceived [by the community] from a height perspective [of number of floors alone]". There is a need for form-based building code digital tools that can assist visualising buildings as "a full package", rather than in simple terms of "height and setbacks", and that could look at the "form of the built environment and how that interface is with the surroundings" and help answer "how does it [a building] actually visually relate to the space of the surround?"

To test the algorithm for form-based codes, an elementary precinct modelling environment was developed in Unity (Figure 3). The environment supported three key precinct entities: building massing and its creation and arrangement; a subject building envelope of parameterised footprint and height; and one or more viewsheds, each oriented towards the subject building envelope and anchored by a user-moveable point. For each viewshed, an in-scene screen shows to the user a view of the subject building. A pattern was applied to the subject building envelope in 3 m bands in the X, Y, and Z axes to highlight its form. A simple user interface was provided to set for the building envelope minimum and maximum allowable heights and a preferred average height (and, by extension, a preferred floor area using an assumed floor-to-floor height of 3 m). The user may also assign a [0,1] importance weighting to each viewpoint.

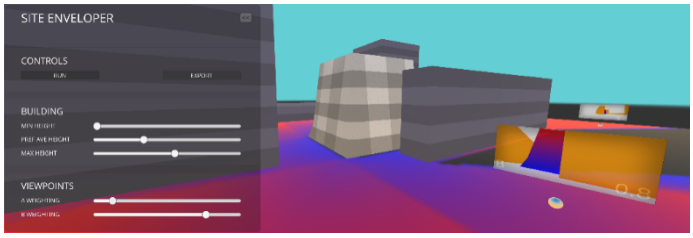


Figure 3. The elementary precinct modelling environment in Unity showing building massing (left rear and mid-right); subject building envelope (centre) and visualisation of its visible height heatmap (ground plane); viewpoints and their screens (right); and a simple user interface (right fore).

A genetic algorithm we developed in C# was applied to generate candidate form-based code solutions for the subject building envelope with respect to the weighted viewsheds (Figure 4). The genetic representation of the subject building envelope for each individual was encoded as a scalar field of heights. The fitness function was formulated to A) minimise the difference of the solution building envelope average height and the user-set preferred average height and B) minimise the weighted visibility of the building envelope from the viewpoints. In lay terms, the envelope optimisation seeks to maintain a desired building volume while hiding that volume from each viewpoint according to its assigned importance insofar as is possible.

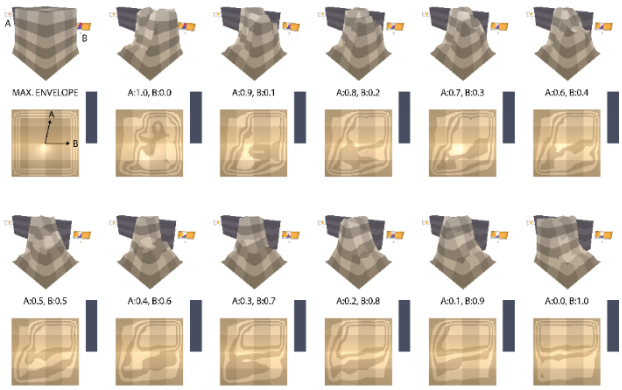


Figure 4. Perspective and orthographic views of 11 candidate solutions for different weightings of two points, A and B, from A: 1.0, B: 0.0 to A: 0.0, B: 1.0 in 0.1 increments. The configuration of building massing is the same as seen in Figure 3.

Performance was tested for solution convergence of the genetic algorithm and computational speed. The tool was tested with a genetic algorithm of population size of 150 individuals each of chromosome length 81 representing a site envelope of an 8 x 8 cell grid. Results indicate the fitness plateaus towards 200 generations (Figure 5). Each generation had a mean calculation time of 901.83 ms (n = 200) on a Windows 10 laptop (i7-8750H, 32GB RAM, NVIDIA GeForce RTX 2070, 1080p). A complex environment was tested by replacing the buildings in the standard scene with massed tree models, thereby increasing the number of triangles the GPU must process for viewpoint A from 2.8k to 673.1k and for viewpoint B from 1.4k to 325.5k. Each generation in this scenario had a mean calculation time of 1,130.4 ms (n = 200).

A traditional hit test method was appraised in the standard scene for comparison. Given the target image used for rendering in the novel approach was 512 x 256 pixels, a minimum of 131,072 hit tests would be needed to achieve the same level of detail. To account for the population size of 150 individuals, each being evaluated for two viewpoints, sets of 39,321,600 building hit tests were undertaken (n = 5) giving a mean time of 58.54 s, and not including any overheads of site envelope geometry manipulation and the genetic algorithm framework.

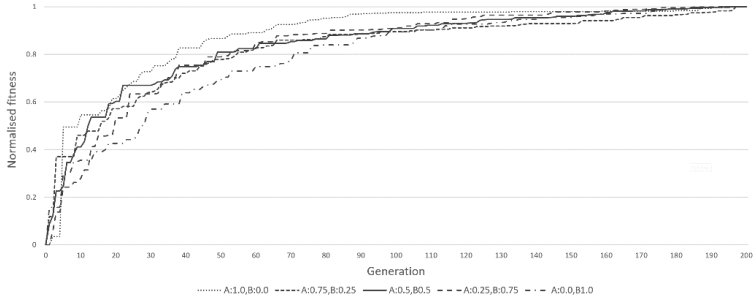


Figure 5. Normalised fitness versus generations for the urban configuration used in Figure 4.

5. Discussion

The experimentation presented in this paper has tested how visuospatial analysis may be embedded and propagated in a design computing workflow. The exploration was conducted with a test case of a real-world industry problem with application of a novel GPU-based visuospatial analysis algorithm.

The GPU shader urban visibility algorithm introduced in the method of this paper exposes visuospatial data for further processing in a design computing workflow. It demonstrated performance approximately 64 times faster than a comparable hit test implementation and has an efficiency that permits interactive application for even complex urban scenes. The level of detail of the analysis can be altered for use case requirements by adjusting the resolution of the render target image. The method does not rely on physics-based intersection testing of 3D geometry and is hence applicable to analysis of 3D models - vegetation, for example - that could otherwise be computationally expensive to test accurately.

Although the algorithm was implemented in this research to analyse building visibility, modifying the processing on the GPU can adapt it to other urban features and metrics, such as vegetation and green view. This study exploited specific features of the Unity 3D game engine graphical system and is extensible to other development platforms like Unreal Engine, and to direct implementation with technologies including OpenGL and DirectX.

The applied test considered the embedding of visuospatial analysis within a design computing workflow and used a platform used in industry software solutions. The building envelopes generated reflected a spatial variance in accord with viewpoint weightings. In the context of form-based building codes developed in collaboration with a local government authority, and community perceptions of the number of floors as a foremost planning and design criteria, the outcomes indicate a possibility for visuospatial digital tools to change the way in which the public is engaged. While the detailed investigation of this possibility is beyond the scope of this research, early indications suggest these tools have the potential to provide the community a more nuanced understanding beyond simple floor count metrics.

Two extensions were considered but not implemented for this research phase. First, linking viewpoint vectors and weightings to ABM pedestrian circulation would be straightforward and would provide a ready basis for each of building and precinct level modelling. Second, envelope generation could accommodate more detailed exploration of design options; for example, mass modelling could permit cantilevers and spatial voids, visibility assessment could consider facade treatments, and additional analysis such as shadowing of public space could be incorporated. In implementing these factors, envelope optimisation would need to be meticulously balanced against rivalrous considerations and practical limitations lest it converges on unworkable outcomes. In this research the optimisation results in recognisable building envelopes due to the assessment of building volume against its visibility and to the inherent constraints of scalar field modelling of envelope height. In further development, a naive implementation could spawn unworkable forms; modelling spatial voids, for instance, may create forms that merely converge on the spatial volume hidden from viewpoints or that poorly allow for structural elements.

Although moving closer to the "full package" of visual analysis desired by the LGA, these two extensions were not pursued in this early exploration as they risk the output solutions becoming difficult to intuitively comprehend or to match to the input conditions and hence less persuasive if used in community engagement. We see this bind as raising the broader question, that appears elsewhere in AI research, of balancing the use of AI outputs that may be black box and not explainable versus sticking to potentially less powerful or less nuanced but more intuitive solutions.

The digital data of view analysis is carried further in the design computing workflow in this research than immediate visualisation, yet ultimately the human is brought back into the design loop. The resulting envelopes of the tool are indicative only and there is no readily definable stopping condition for the algorithm that is optimal in objective terms. Each envelope is not prescriptive and must be interpreted by the designer or planner as spatial bounds to observe or break according to their nuanced understanding of urban environments. Further research could study the utility and interpretation of generated envelopes and how far such data can or should flow in a design computing workflow - for the human must return to the design loop if we are not considering a fully automated design machine with all its innate social and technical challenges.

In our research, we have introduced a novel analysis algorithm to investigate a gap between the potential to further propagate visuospatial data within a design computing workflow and the functionality offered by typical software tools available to urban built environment professionals. In the context of form-based codes, the results demonstrate the human within the design computing loop can benefit from extension of computation on visuospatial analysis outputs, and that this strategy can be feasibly employed by developers of industry software solutions.

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